time, or having multiple answers. The questions with violations detected by ChatGPT were sent back to AI trainers for revision. At the end of the question creation stage, we used ChatGPT to lightly re-write questions to improve grammar and punctuation without modifying the content of the question.

In the verification stage of this dataset, each question was answered by an independent AI trainer without access to the answer given by the question creator. In this stage we also had AI trainers answer yes/no questions for whether questions only had single, indisputable answers, and whether answers to questions would stay the same over time.

After the verification stage, we removed questions from the dataset if answers from two trainers didn't agree according to a prompted ChatGPT classifier, or if the second trainer said the question wasn't timeless or didn't have an indisputable answer. To improve the correctness of reference answers, we also only kept questions for which there were two unique website domain names among the 2–4 sources found by the two AI trainers (one source from the first AI trainer, and 1–3 from the second AI trainer).

After the dataset was finalized with reference answers agreed on by two trainers, we did an additional quality check by randomly selecting 1,000 examples and having them answered by a third trainer. According to the prompted ChatGPT grader, the performance of the third trainer was 94.4%. Of the 5.6% of answers (56/1000) classified as incorrect by the prompted ChatGPT grader, we did a manual examination of all examples and found that fifteen were false negatives from the autograder. Of the remaining 41 (4.1%) actually incorrect answers, we found that seven were due to the third trainer not answering the question fully (e.g., giving only the year when the question asks for month and year), and six were due to the third trainer giving an answer that contradicted the source they cited (e.g., they misread the own source they cited).

The remaining 2.8% of the errors from the third trainer revealed real issues with the data. Hence, we estimate that the error rate of our benchmark is around 3%, assuming no false positives from the prompted ChatGPT grader. The most common issues were ambiguous questions (e.g., not specifying driving vs flying distance between cities), reputable sources giving contradictory information (e.g., history.com and wikipedia.com give different dates for when Nixon retired from the US Naval Reserve), and having more than one correct answer (e.g., John Lennon's psychedelic Rolls-Royce was shown at the Pacific National Exhibition in two years: 2014 and 2015).

2.3 Dataset diversity

SimpleQA contains questions from a diverse range of topics, which we tagged post-hoc with ChatGPT. The most common topics were Science & Technology (n=858), Politics (n=709), and Art (n=550). Figure 1 shows the proportion of each topic in a pie chart.

In addition to question topic, we can also look at the diversity of data in a few other axes. Using ChatGPT to classify types of answers, we found that 32.8% of answers were dates, 24.1% of answers were a person, 15.3% answers were a number, 9.9% answers were a place, and 18.0% answers were classified as "other." As for diversity in sources, we see that `wikipedia.com` is by far the biggest source (one of the sources for 3.5k of 4.3k questions), followed by `fandom.com` (410 questions), `ac.uk` (154 questions), and `imdb.com` (121 questions).

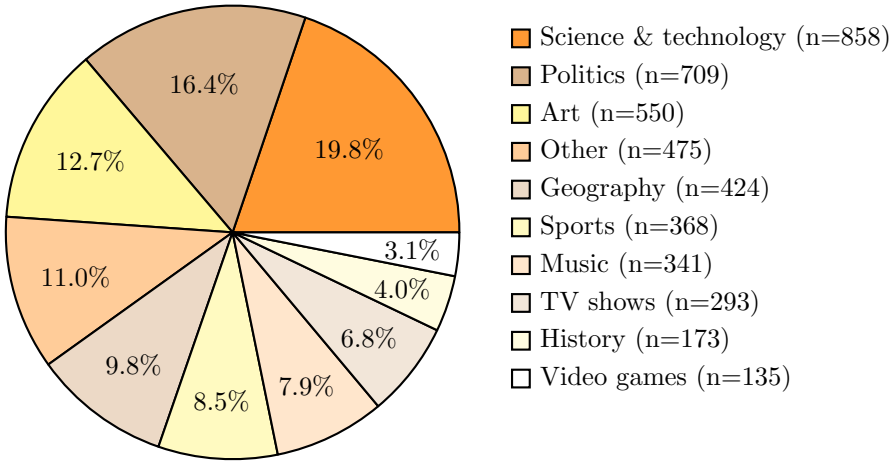


Figure 1: Distribution of topics in SimpleQA. The topic for each question was classified via a prompted ChatGPT model.

2.4 Grading and metrics

To grade completions, we use a prompted ChatGPT classifier that sees both the predicted answer and the reference answer, and grades responses as either "correct," "incorrect," or "not attempted." The definitions for each of these grades, along with a few example responses, are shown below in Table 2.

Grade	**Definition**	**Example responses**
Correct	The predicted answer fully contains the reference answer without contradicting the reference answer.	"Wout Weghorst", "Wout Weghorst scored at 83' and 90+11' in that game"
Incorrect	The predicted answer contradicts the reference answer in any way, even if the contradiction is hedged.	"Virgil van Dijk", "Virgil van Dijk and Wout Weghorst", "Wout Weghorst and I think van Dijk scored, but I am not totally sure"
Not attempted	The reference answer is not fully given in the answer, and there are no contradictions with the reference answer.	"I don't know the answer to that question", "To find which Dutch player scored in that game, please browse the internet yourself"

Table 2: Grading categories with examples completions. The question here is "Which Dutch player scored an open-play goal in the 2022 Netherlands vs Argentina game in the men's FIFA World Cup?" (Answer: Wout Weghorst).

You can view the full prompt to the grader in Appendix A. We did not do a formal study of the performance of the grader but in practice, we found that it works pretty well. Of 100 correct, 100 incorrect, and 100 not attempted completions we manually read, we only found two disagreements with the prompted grader.

For any evaluation benchmark, having a single-number metric can be of great utility, even if it is imperfect. One way to do this is to first summarize "correct", "incorrect", or "not attempted" into two metrics that can be thought of as similar to (but not exactly the same as) recall and precision:

- A metric called *overall correct* (or just "correct") is simply what percent of all questions were answered correctly.
- A metric called *correct given attempted* is what percent of questions the model answered correctly, out of only questions that were attempted (i.e., questions answered correct and incorrectly).

To get a single-number metric, we can compute an F-score as the harmonic mean of overall correct and correct given attempted. To give a sense of how F-score captures precision and recall in this case, a model that always attempts to answer and gets 30% correct would get an F-score of 30%; a model that has correct-given-attempted of 80% would only need to get overall-correct of 19% to get an F-score of 30%. However, an issue with F-score is that if model performance is below 50%, it always makes sense for the model to try to guess if it is at least 50% sure that it will get an answer correct (see Appendix B for why this is the case; Kalai (2024)).

A single-number metric that does not have a loophole would be to assign a specific negative penalty p to wrong answers, and then simply to take the average score where correct is worth 1 point, not attempted is worth 0 points, and incorrect is worth $-p$ points. This metric is useful if one is willing to set a somewhat arbitrary threshold for $-p$ for their particular use case. This metric can be interpreted as how much better a model is at getting answers correct than incorrect, with respect to the threshold. At $p = 9$, the weighted sum of the scores for a model would only be positive if the model was getting at least 90% of the problems it attempted correct (a bar that none of the models we evaluate in this paper currently meet).

3 Evaluation of models

As shown in shown in Table 3, we evaluated various OpenAI (OpenAI, 2024a,b,c) and Anthropic models (Anthropic, 2024) on SimpleQA. As expected, we see that larger models have higher performance than smaller models (GPT-4o outperforms GPT-4o-mini; o1-preview outperforms o1-mini and opus is the highest performance of the claude-3 series).

Because we created questions to be hard for GPT-4o, looking at the performance of Claude is a good sanity check of whether the process of creating hard questions for GPT-4o resulted in a dataset that was only hard for GPT-4o but easy for other models. We see that Claude's performance is also not super high, so it is likely that SimpleQA is a challenging dataset for frontier models generally. Another interesting observation with the Claude models is that they tend to not attempt questions more often than the GPT-4o models. For instance, Claude-3.5 Sonnet has much fewer correct questions than GPT-4o, but also attempts much fewer questions, resulting in a similar F-score.